Ex

$$\mathcal{A} \in \mathbb{R}^{1\times2\times3}$$

$$\mathcal{A}[:,:,1] = [1 \quad 0]$$

$$\mathcal{A}[:,:,2] = [1 \quad -1]$$

$$\mathcal{A}[:,:,3] = [0 \quad 1]$$

i.e $\mathcal{A} = [[1 \quad 0], [1 \quad -1], [0 \quad 1]]$

Step I unfold $\mathcal{A}$ along mode - 1, 2, 3

a) unfold along mode - 1

$$\mathcal{A}_{(1)} = [1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 1]$$

b) unfold along mode - 2

$$\mathcal{A}_{(2)} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

c) unfold along mode - 3

$$\mathcal{A}_{(3)} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Step II Find left singular vectors of $\mathcal{A}_{(i)}$

i.e eigenvectors of $\mathcal{A}_{(i)}\mathcal{A}_{(i)}^T$

a) Construction of U_1.

$$\mathcal{A}_{(1)}\mathcal{A}_{(1)}^T = [1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 1] \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$= [4]$$

eigenvalue is 4 and eigenvector is $\boxed{[1] = U_1}$

b) Construction of U_2

$$\mathcal{A}_{(2)} \mathcal{A}_{(2)}^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

eigenvalues are $\lambda^2 - 4\lambda + 3 = 0$

$\lambda = 3, 1.$

eigenvectors are $\left[\begin{array}{cc|c} -1 & -1 & 0 \\ -1 & -1 & 0 \end{array}\right] \rightsquigarrow v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ $v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$U_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

b) Construction of U_3

$$\mathcal{A}_{(3)} \mathcal{A}_{(3)}^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

eigenvalues are $\lambda = 3, 1, 0$

Eigenvectors are $\left\{ \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right\}$

$$U_3 = \begin{bmatrix} -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}$$

Step III Construction of S.

$$S = \mathcal{A} \times_1 U_1^T \times_2 U_2^T \times_3 U_3^T$$

Consider

$$\mathcal{A} \times_1 U_1^T = \text{fold}_1 (U_1^T \text{unfold}_1 \mathcal{A})$$

$$= \text{fold}_1 ([1][1 \ 0 \ 1 \ -1 \ 0 \ 1])$$

$$= [[1 \ 0] \ [1 \ -1] \ [0 \ 1]] = \mathcal{B}$$

$$\mathcal{B} \times_2 U_2^T = \text{fold}_2 (U_2^T \text{unfold}_2 \mathcal{B})$$

$$= \text{fold}_2 \left(\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \right)$$

$$= \text{fold}_2 \left(\begin{bmatrix} 1/\sqrt{2} & 2/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \right)$$

$$= [[1/\sqrt{2} \ \ 1/\sqrt{2}] \ [2/\sqrt{2} \ \ 0] \ [-1/\sqrt{2} \ \ 1/\sqrt{2}]] = \mathcal{C}$$

$$\mathcal{C} \times_3 U_3^T = \text{fold}_3 (U_3^T \text{unfold}_3 \mathcal{C})$$

project the data onto this eigenvector.

$$= \mathrm{fold}_3\left(\begin{pmatrix} -1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}_{3\times 3} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 2/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}_{3\times 2} \right)$$

$$= \mathrm{fold}_3 \begin{pmatrix} -\sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$= [[\sqrt{3} \quad 0] \; [0 \quad 1] \; [0 \quad 0]]$$